



Supervised Relevance Analysis for Multiple Stein Kernels for Spatio-Spectral Component Selection in BCI Discrimination Tasks

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Abstract. Brain-Computer Interfaces (BCI) allow direct communication from the human brain to machines by analyzing sensorimotor activity. Within the most common paradigms of cognitive neuroscience for BCI, the Motor Imagery one relies on imaging movement of hands, feet, and tongue, usually. Standard BCI systems make use of electroencephalographic signals due to its high temporal resolution, portability, and easiness to implement. Recently, the filter-banked analysis of common spatial patterns has proved to be among the most discriminative approaches. Nonetheless, such an approach results in high dimensional spaces yielding to overtrained systems, or ill-posed spatial covariance matrices, when the number of EEG sensors increases. In this work, we propose a spatio-spectral relevance analysis, termed Multiple Stein Kernels (MSK), to simultaneously select channels and bands in a supervised criterion, so reducing the spatial and frequency dimension. We test our approach in three known BCI datasets with a varying number of subjects and channels. The attained classification results prove that MKS not only improves the classification accuracy and reduces the dimension of the representation space, but also select frequency bands that are physiologically interpretable.

1 Introduction

Nowadays, there are many challenges in rehabilitation in the development of systems that enhance communication between the brain and the exterior environment. Conventional Brain-Computer Interfaces (BCI) have been widely used to assist disabled people to re-establish their capabilities of environmental control decoding the brain activity [7], being the electroencephalography (EEG) signal the most commonly used acquisition technique due to its implementation easiness and noninvasive implementation [4].

Among the BCI systems, the Motor Imagery (MI), corresponding to imagining a motor action without execution, is the most known paradigm. Feature extraction approaches, as common spatial patterns (CSP), are mainly devoted to MI classification tasks by reducing the characteristic noise, while highlighting mental imagination patterns. Further, the spectral CSP extensions try to find the frequency band with the highest performance focusing on the relationship between CSP features and the frequency. For instance, Sparse Filter Band Common Spatial Pattern (SFBCSP) performs a sparse regression to select the most discriminant bands from the full channel set [8]. Nonetheless, CSP-based approaches demand a linear projection of channels that hinders non-linear data distributions [6]. Recently, the Riemannian geometry has improved the performance of conventional BCI algorithms relying on the EEG covariance as the feature representation, so avoiding any linearity assumption. In fact, spatial covariance matrices carry useful information for MI, namely, their diagonal retrieves the sensor power, and the off-diagonal entries give extra information about channel correlations. However, in the case of EEG-based MI classification, the Riemannian geometry suffers from the curse of dimensionality. That is, the number of sensors hampers the classification performance, due to more samples are needed to build non-singular covariance matrices. When nearly singular covariance matrices are produced, their ill-posing cannot be efficiently handled via Riemannian geometry [5].

For overcoming the above issues, we propose a spatio-spectral relevance analysis simultaneously selecting channels and bands in a supervised scheme. The approach, termed Multiple Stein Kernel (MSK), reduces the spatial and frequency dimension by decreasing the chances of an EEG recording to result in an ill-posed covariance matrix. Then, the Stein kernel, that assesses the inner-product for covariance matrix spaces overcomes feature extraction stages. Since the Stein kernel is applied for each frequency band, a multiple kernel learning strategy provides a single similarity measure to feed classification machines. Attained results prove that the introduced relevance analysis improves the classification performance, and enhances the representation of the MI paradigm while reducing the spatio-spectral components.

2 Materials and Methods

2.1 EEG Filter-Bank Decomposition

Let $\mathbf{X} = \{\mathbf{x}_n^c \in \mathbb{R}^T : n \in N, c \in C\}$ be a set of N acquired EEG recordings of length T along C channels, where each recording is provided with a label $l_n \in \{-1, +1\}$. Each recording is further passed through a bank of B linearly distributed filters of elemental bandwidth $\Delta_\omega \in \mathbb{R}^+$ and overlap $\delta_\omega \in \mathbb{R}^+$. Thus, the set of bandpass-filtered EEG data is obtained, $\mathbf{X} = \{\mathbf{x}_{nb}^c \in \mathbb{R}^T : b \in B\}$. As recommended in the literature, we consider $B = 17$ band-pass filters with a bandwidth of $\Delta_\omega = 4$ Hz aiming to cover the whole relevant EEG frequencies, ranging from 4 to 40 Hz and fixing the overlap between each other at $\delta_\omega = 2$ Hz.

2.2 Proposed Relevance Analysis

To account the relevance of a single channel at each frequency band, we build a relevance index based on the cross-correlation operator $R\{\cdot, \cdot\} \in [-1, 1]$ and the Centered-Kernel Alignment measure. We compute the cross-correlation function between all record pairs for each band and channel following:

$$R_\tau\{\mathbf{x}_{nb}^c, \mathbf{x}_{mb}^c\} = \frac{\mathbb{E}_t \{x_{nb}^c(t)x_{mb}^c(t-\tau)\}}{\sqrt{\mathbb{E}_t \{(x_{nb}^c(t))^2\} \mathbb{E}_t \{(x_{mb}^c(t-\tau))^2\}}}; \forall b \in [1, B], c \in [1, C] \quad (1)$$

where $\mathbb{E}_z \{v(z)\}$ stands for the expectation of the vector v across the z axis and $x(t)$ denotes the EEG measure at time instant t . From the cross-correlation function we build a symmetric matrix $\mathbf{I}_b^c \in [0, 1]^{N \times N}$ with elements $\gamma_b^c(n, m)$:

$$\gamma_b^c(n, m) = \max_\tau |R_\tau\{\mathbf{x}_{nb}^c, \mathbf{x}_{mb}^c\}| \in [0, 1]. \quad (2)$$

Relying on the properties of the cross-correlation operator, the symmetric matrix $\mathbf{I}_b^c$ is also positive definite, allowing to account for the matrix similarity using the kernel-devoted metrics. Particularly, we assess the centered-kernel alignment (CKA) $\rho_b^c \in [0, 1]$ between $\mathbf{I}_b^c$ and $\mathbf{L} \in \mathbb{R}^{N \times N}$, that holds elements $L_{nm} = \delta(l_n - l_m)$ and accounts for the supervised information:

$$\rho_b^c = \frac{\langle \bar{\mathbf{I}}_b^c, \bar{\mathbf{L}} \rangle_F}{\sqrt{\|\bar{\mathbf{I}}_b^c\|_F \|\bar{\mathbf{L}}\|_F}}; \forall b \in [1, B], c \in [1, C], \quad (3)$$

where $\bar{\mathbf{L}}$ stand for the centered version of $\mathbf{L}$. Notations $\|\cdot\|_F$ and $\langle \cdot, \cdot \rangle_F$ correspond to the Frobenius norm and dot product, respectively. As a result, ρ_b^c measures the similarity between EEG data and their labels at spatial and frequency levels, allowing to rank each channel and band according its discriminative capacity.

2.3 Classification Based on Stein Kernel Representation

The channels of each band are ranked according the score given by the CKA criterion, in this way the most relevant channels are selected for each band, so that the spatial covariance only holds the most discriminative information as $\mathbf{\Sigma}_{nb} = \tilde{\mathbf{X}}_{nb} \tilde{\mathbf{X}}_{nb}^\top$, being $\tilde{\mathbf{X}}_{nb} \in \mathbb{R}^{C' \times T}$ corresponds to the n -th EEG recording at band b with $C' \geq C$ selected channels. Covariance matrices belong to a special Riemannian manifold in the space of symmetric matrices, that can be endowed with dissimilarity metrics to estimate the matrix distances. Here, we consider the Stein divergence, also known as LogDet Divergence, that defines the not geodesic distance between two symmetric positive definite (SPD) matrices $\mathbf{\Sigma}_n$ and $\mathbf{\Sigma}_m$ as follows:

$$d_{SD}^2(\mathbf{\Sigma}_{nb}, \mathbf{\Sigma}_{mb}) = \log \det \left(\frac{\mathbf{\Sigma}_{nb} + \mathbf{\Sigma}_{mb}}{2} \right) - \log \det \left(\frac{\mathbf{\Sigma}_{nb} \mathbf{\Sigma}_{mb}}{2} \right) \quad (4)$$

In the proposed methodology, we feed the Stein divergence into a radial basis function to build a Gaussian Kernel on the Riemannian geometry as:

$$\kappa(\tilde{\mathbf{X}}_{nb}, \tilde{\mathbf{X}}_{mb}) = \exp(-d_{SD}^2(\mathbf{\Sigma}_{nb}, \mathbf{\Sigma}_{mb})/2\sigma^2) \quad (5)$$

where the kernel $\kappa(\cdot, \cdot) \in [0, 1]$ corresponds to the Stein kernel and $\sigma \in \mathbb{R}^+$ to its bandwidth. Since each selected band b' yields its own similarity, we introduce the multiple kernel learning to gather all similarity functions:

$$\kappa(\mathbf{X}_n, \mathbf{X}_m) = \sum_{b'=1}^{B'} w_{b'} \kappa(\tilde{\mathbf{X}}_{nb'}, \tilde{\mathbf{X}}_{mb'}) \quad (6)$$

where $w_b' \geq 0$ holds the mixture weights that must be optimized to improve discrimination between classes based on the label set. By constraining the weights to sum the unit, $\sum_{b'} w_{b'} = 1$, yields a convex function that has a closed form solution that maximizes the CKA criterion. Therefore, we use a support vector machine equipped with the proposed Multiple Stein Kernel (MSK) for covariance matrices. As a result, the proposed approach reduces the spatio-spectral components based on the relevance analysis, with the additional benefit of simplifying the BCI system as the explicit feature extraction stage is avoided.

3 Experimental Set-Up

3.1 Dataset Description

Being the most used MI paradigm in the literature, this work considers the binary classification of left and right movement for three datasets:

BCICIII4a: The BCI Competition III dataset IVa¹ contains the EEG signals recorded from five subjects performing two different (right hand and left hand) MI tasks. 118 channels at the positions of the extended international 10/20 system were used for measuring the EEG signal, sampled at a rate of 1000 Hz. The dataset contains a total of 280 trials for each subject with an equal number of trials for each task. The data provided were interval filtered with a passband of 0.05–200 Hz the down-sampled data at 100 Hz is used. Visual cues, lasting 3.5 s indicated which of the following three motor imageries the subject should perform: (L) left hand, (R) right hand, (F) right foot. The presentation of target cues was intermitted by periods of random length, 1.75 to 2.25 s, in which the subject could relax.

BCICIV2a: The BCI competition IV dataset IIa² contains a collection of EEG signals recorded using a 22-electrode montage from nine subjects that compose the dataset. Each subject performs two sessions on different days with four motor

¹ <http://www.bbc.de/competition/iii/>.

² <http://www.bbc.de/competition/iv/>.

imagery tasks. Namely, left hand, right hand, both feet, and tongue on two sessions. The session includes six runs with twelve trials per task, obtaining 144 trials for each class. The signals were sampled with 250 Hz and bandpass-filtered between 0.5 Hz and 100 Hz. The subjects were sitting in a comfortable armchair in front of a computer screen. At the beginning of a trial ($t = 0$ s), a fixation cross appeared on the black screen. Also, a short acoustic warning tone was presented. After two seconds ($t = 2$ s), a cue in the form of an arrow pointing either to the left, right, down or up (corresponding to one of the four classes left hand, right hand, foot or tongue) appeared and stayed on the screen for 1.25 s. This fact prompted the subjects to perform the desired motor imagery task. No feedback was provided. The subjects were asked to carry out the motor imagery task until the fixation cross disappeared from the screen at $t = 6$ s.

GIGASCIENCE: The GIGASCIENCE³ dataset contains an EEG signal recorded conducting a BCI experiment for the MI movement of the left and right hands with 52 subjects [1] of which we selected 44 subject for the current experiment because the remainder subject holds a large number of trials with the artifact. EEG data were collected using 64 Ag/AgCl active electrodes 64-channel montage based on the international 10-10 system of sampling rate 512 Hz. The dataset contains 100 or 120 trials for each MI class. Each trial starts with the monitor showed a black screen with a fixation cross for $t = 2$ s then one of two instructions (“left hand” or “right hand”) appeared randomly on the screen for $t = 3$ s, and subjects were asked to move the appropriate hand depending on the instruction given. After the movement, when the blank screen reappeared, the subject was given a break for a random, ranging from 4.1 to 4.8 s. These procedures were repeated 20 times for one class (one run), and one run was performed.

3.2 Parameter Tuning

The proposed method MKS requires two procedures of parameter tuning, namely, the kernel bandwidth $\sigma \in \mathbb{R}^+$ and the threshold $\lambda \in (0, 1)$ controlling the amount of selected spatio-spectral components. The parameter σ and λ are tuned based on a cross-validation grid search for the best classification accuracy. The spatio-spectral components used for the classification are selected according to the relevance analysis ranking. Figure 1 illustrates the learning curve for each subject in the BCICIV IIa dataset, marking the best number of components. Note that the maximum accuracy for each subject is achieved at a different amount of components, evidencing the need for a subject-wise tuning of λ to properly encode the discriminant brain activity.

Further, observing the learning curves, we identify two kinds of learning, one for subjects with high performance (S1, S3, S5, S7, S8, S9) and another that embraces all subjects with lower performance (S2, S4, S6). In the first kind, the accuracy monotonously grows and stabilizes at a few components, so

³ <http://gigadb.org/dataset/100295>.

revealing the goodness of our approach for the selection of relevant components. In the second kind, there are abrupt changes induced by noisy components that disturb the classifier. However, the proposed relevance allowed to select a subset of components outperforming the full component set.

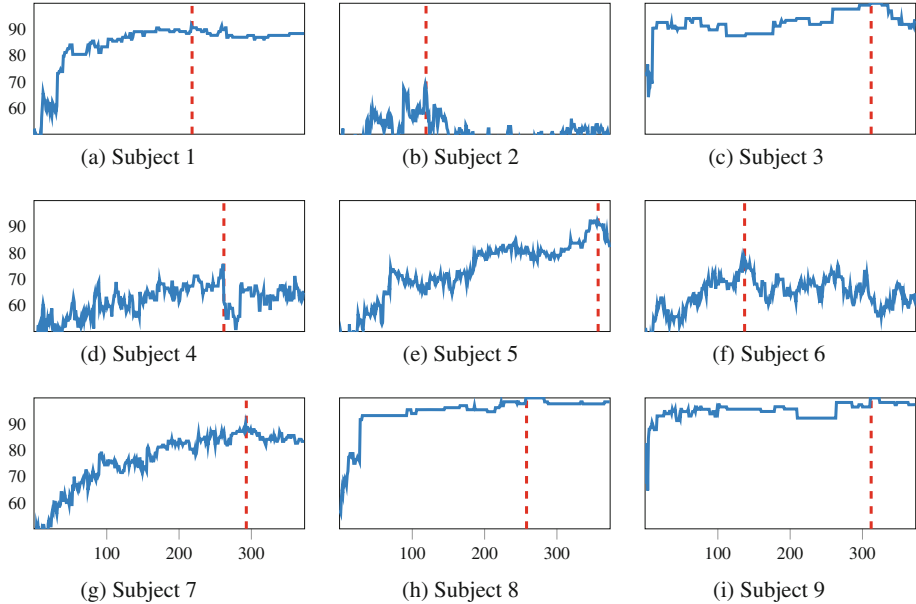


Fig. 1. BCICIV2a learning curves. Accuracy versus the number of selected components. The best accuracy is marked by a red dashed line for each subject. (Color figure online)

3.3 Performance Results

We compare the MKS against the conventional CSP, Sparse Filter-bank (SFB) CSP [8], and the Stein kernel on a single band [3] for the considered datasets. Fivefold cross-validation is used to evaluate the performance of all experiments conducted using BCICIII4a, BCICIV2a, and GIGASCIENCE datasets. Table 1 summarizes the accuracy performed by each contrasted approach along with the standard deviation, the component reduction rate (*crr*), and the *p*-value of the significance test of the proposed approach against each competing one. Results evidence that MKS outperforms the baselines at all considered performance metrics for the three datasets. Particularly, our proposed approach obtains better accuracy, using fewer components that could be redundant or noisy for the MI task. Besides, obtaining a lower standard deviation proves that MKS also provides more confidence than the approaches of the state-of-the-art do.

Figure 2, showing the most selected bands per channel on each considered database, indicates that MKS suitably incorporates frequency information on

Table 1. Classification accuracy for the considered approaches on a five-fold cross-validation.

	CSP	Stein kernel	SFB	MKS
<i>BCICIII IVa</i>				
$\mu \pm \sigma$	87.6 ± 4.2	84.3 ± 4.4	87.7 ± 2.4	92.4 ± 3.1
p-value	$2.0e^{-3}$	$8.8e^{-6}$	$5.1e^{-3}$	–
crr	118/118	2006/2006	2006/2006	867/2006
<i>BCICIV IIa</i>				
$\mu \pm \sigma$	76.5 ± 5.6	79.8 ± 7.0	84.5 ± 5.4	88.6 ± 5.1
p-value	$4.8e^{-7}$	$2.1e^{-4}$	$6.6e^{-3}$	–
crr	22/22	374/374	374/374	252/374
<i>GIGASCIENCE</i>				
$\mu \pm \sigma$	68.9 ± 7.6	70.8 ± 6.6	72.2 ± 6.3	78.1 ± 6.5
p-value	$4.0e^{-7}$	$7.9e^{-28}$	$4.1e^{-13}$	–
crr	64/64	1088/1088	1088/1088	446/1088

specific bands that encode MI related brain activity. Notably, the most discriminating bands are 12–16 Hz, 14–16 Hz, and 8–10 Hz for BCICIII4a, BCICIV2a, and GIGASCIENCE, respectively. All these frequencies are strongly related to the MI task because finding that imagination of movement leads to short-lasting and circumscribed attenuation (or accentuation) in mu (8–12 Hz) and beta (13–28 Hz) rhythms [2].

4 Concluding Remarks and Future Work

This work proposes a spatio-spectral relevance analysis to select the most discriminating spatial bandpass filtered components of EEG recordings, termed MKS. As the accuracy results prove, the supervised relevance criterion information enhances the stability and the discriminative capability in the representation space. That is the case of the Stein kernel along with the Riemannian geometry that suffers from stability problems at large dimensionality, that in our application results from the number of sensors defining the size of covariance matrices. The proposed approach deals with such an issue by reducing the representation dimension. Also, MKS incorporates frequency information allowing the selection of bands that encode the information related to brain activity, as shown in Fig. 2. Attained results prove that the relevant information is found in bands 12–16 Hz, 14–16 Hz, and 8–10 Hz for BCICIII4a, BCICIV2a, and GIGASCIENCE, respectively. All those bands are strongly related to the MI task because the movement imagination leads to short-lasting and circumscribed attenuation (or accentuation) in μ and β rhythms. Including the above relevant frequency information in the feature selection stage improves the accuracy on subjects such as S5 and S2 of BCICIV2a. Therefore, our proposed approach obtains a better

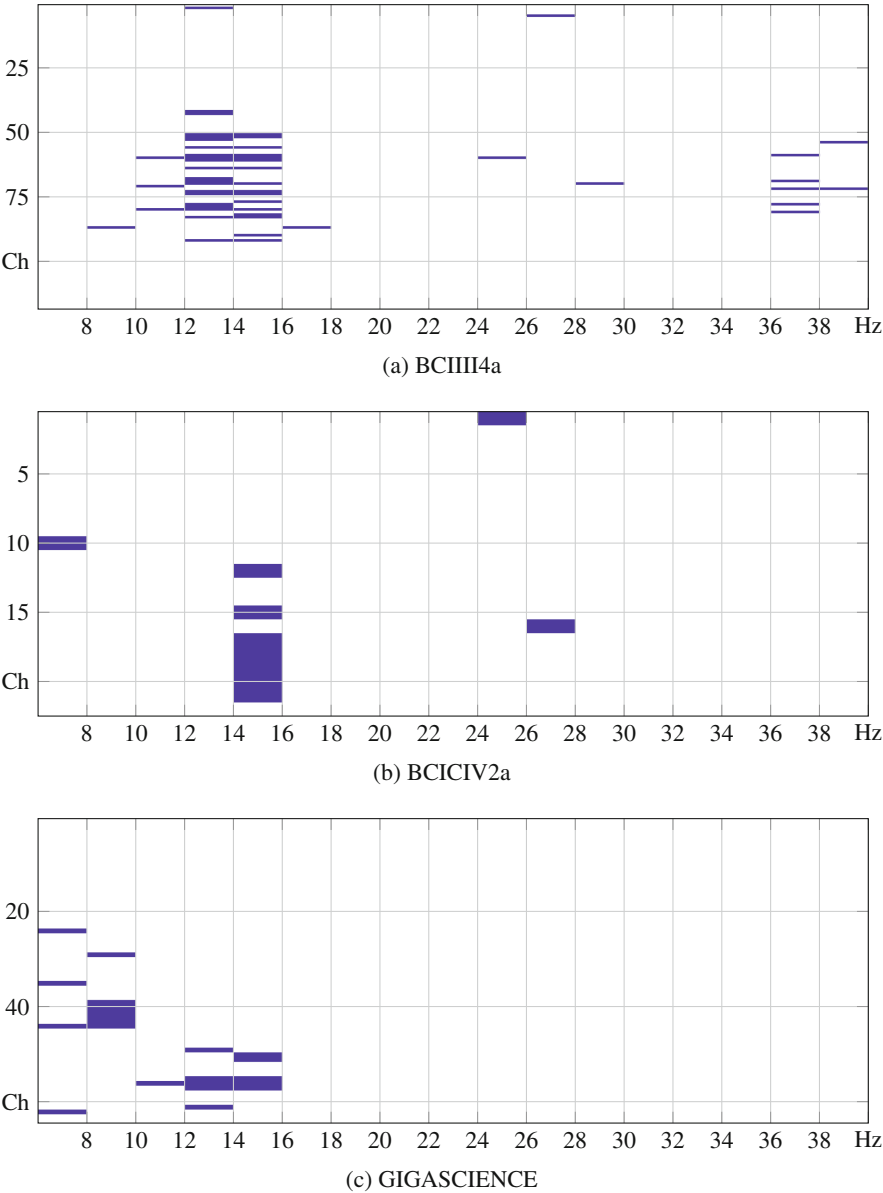


Fig. 2. The most selected frequency bands and EEG channels for each dataset. Y-axis represents the channels distributed from frontal (top) to occipital (bottom). X-axis indicates the center of the frequency band.

accuracy compared with the other methods evaluated for the three BCI datasets, using either fewer channels or frequency bands, by eliminating noisy components for the MI task. As an additional benefit, MKS attains lower standard deviations, yielding to a more confident approach in comparison with other state-of-the-art-techniques in the statistical sense.

As future work, the authors plan to develop an optimization approach allowing to map covariance to PSD matrix spaces with fewer components that cope with the redundant information. In this sense, both the channel and frequency feature extraction will be carried out in a unified framework that results in globally optimal setups. Also, another EEG-based classification tasks such as disease diagnosis support will be considered to test the proposal on different scenarios.

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References

1. Cho, H., Ahn, M., Ahn, S., Kwon, M., Jun, S.C.: EEG datasets for motor imagery brain-computer interface. *GigaScience* **6**(7), 1–8 (2017)
2. Huang, G., Liu, G., Zhang, D., Zhu, X.: Model based generalization analysis of common spatial pattern in brain computer interfaces. *Cogn. Neurodyn.* **4**(3), 217–223 (2010)
3. Nguyen, C.H., Artemiadis, P.: EEG feature descriptors and discriminant analysis under Riemannian manifold perspective. *Neurocomputing* **275**, 1871–1883 (2018)
4. Ortiz-Rosario, A., Adeli, H.: Brain-computer interface technologies: from signal to action. *Rev. Neurosci.* **24**(5), 537–552 (2013)
5. Yger, F., Berar, M., Lotte, F.: Riemannian approaches in brain-computer interfaces: a review. *IEEE Trans. Neural Syst. Rehabil. Eng.* **25**(10), 1753–1762 (2017)
6. Yger, F., Lotte, F., Sugiyama, M.: Averaging covariance matrices for EEG signal classification based on the CSP: an empirical study. In: 2015 23rd European Signal Processing Conference (EUSIPCO), pp. 2721–2725 (2015)
7. Zhang, Y., Wang, Y., Jin, J., Wang, X.: Sparse Bayesian learning for obtaining sparsity of EEG frequency bands based feature vectors in motor imagery classification. *Int. J. Neural Syst.* **27**(02), 1650032 (2017)
8. Zhang, Y., Zhou, G., Jin, J., Wang, X., Cichocki, A.: Optimizing spatial patterns with sparse filter bands for motor-imagery based brain-computer interface. *J. Neurosci. Methods* **255**, 85–91 (2015)